Computer Vision

Homework 4: Computer Vision and Deep Learning

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In the spirit of Assignment 4 of CS 585 (Spring 2024), a programming assignment on computer vision and deep learning. Goes with Chapter 4 of the notes. You will need PyTorch on your own computer or on a cloud notebook.

1. **A network by hand.** Extend the NumPy training loop of Section 4.1.3 with a convolution layer (use your `conv_layer` from Exercise 4.1.1 for the forward pass and derive the backward pass yourself). Check your gradients with finite differences, then train it on the bar images and compare its accuracy and number of parameters with the fully connected network.
2. **Transfer learning.** Pick a small image classification problem of your own with 3 to 10 classes and at least 50 images per class (your own photos, or a public dataset). Fine-tune a network pretrained on ImageNet, such as a ResNet-18 from `torchvision`, first retraining only the last layer and then fine-tuning everything with a small learning rate. Report the accuracy on a held-out test set and the confusion matrix, and show some correctly and some wrongly classified images.
3. **Verification.** Use the pretrained network, without its last layer, as an embedding: compute cosine similarities between all pairs of test images, treat pairs from the same class as genuine and pairs from different classes as impostors, and plot the ROC curve and the equal error rate as in Section 4.2.2. How does it change after fine-tuning?
4. **Discussion.** Which of your classes are confused, and does that make sense from the pictures? What would you change about your data to do better? Who could be affected if a system like yours were deployed, and how (Section 5.8)?

Hand in your code or notebook and a report (PDF) with the results.